



University of Colombo, Sri Lanka

University of Colombo School of Computing



BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

First Year Examination - Semester I - UCSC AY22 [held in September/October 2025]

IS1207 - Internet and Web Technologies

(Two (2) Hours)

Answer ALL questions

Number of Pages = 12

Number of Questions = 4

To be completed by the candidate

Index Number:

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8

Important Instructions to candidates:

- I. Students should answer in the medium of English language only using the space provided in this question paper.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number **CLEARLY** on each and every page of this Question paper.
- IV. This paper consists of 4 questions in 12 pages (including the Cover Page).
- V. Answer **ALL** questions.
- VI. Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.
- VII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate

To be completed by the examiners

1	
2	
3	
4	
Total	

Question 1

State whether the following HTML, and CSS code segments are syntactically **correct or wrong**. If they are wrong, then make necessary changes to correct them.

[I to XII:2x12=24 Marks and XIII=1Mark]

I. `<head> <link rel="icon" type="image/x-icon" href="/images/favicon.ico"> </head>`

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II. `<body> <meta name="description" content="Free Web tutorials" /> <meta name="keywords" content="HTML, CSS, XML, JavaScript" /> <meta name="author" content="perera" /> </body>`

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III. `<!doctype html public "-//w3c//dtd xhtml 1.0 transitional//en" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"> <html lang="en" xml:lang="en"> - </html>`

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IV. `<body> <a src="applicationform.docx"> Application (doc) </a> <link href="applicationform.pdf">PDF </a> </body>`

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V. `<fig> <figure>Fig.1 - Sri Lanka.</figcaption> </figure>`

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VI. `<body>
habitats
food

<iframe name="abc" embed="habitat.html" width=300
height=200> </iframe> </body>`

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VII. `<head><style> * {background-color: Fuchsia; border-style:
double; border-color: Lime;} </style> </head>`

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VIII. `li a:hover,.dropdown:hover.dropbtn {background-color: red;}
td:nth-child(even), th:nth-child(even) {background-color:
#D6EEEE;}`

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IX. `<head><script>document.create("pal")</script>
<style type="javascript/css">pal{border-style:double;}
</style> </head><body> <h1>A Heading</h1> <pal>My Hero
</pal> </body> </html>`

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X. `<video width="320" height="240" menu>
<source href="abc.mp4" type="mp4"> </video>
<body bgcolor="#2233ad" bgsound src="sound.wav"> </body>`

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XI.
 <map name="can"> <area shape="rect" coords="8,38,87,63"
 href="comet01.html" alt=xxx>
 <area shape="circle" coords="8,38,20" href="comet02.html"
 alt=yyy> </map>

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XII. <!--in the page1-->
 <body> Part A
 Part B
 Part C
 <h1> Part A</h1> <p>Part A.
 Top </p> <h1> </body>
 <!--in the page2-->
 <body>Part C</h1> <p>Part C. </p> </body>

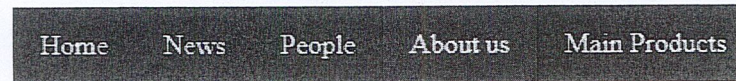
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XIII. @for \$ifrom 1px to 3px{.border-#{i}{border:\$isolid blue;}}

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Question 2

- I. Based on the browser menu output shown below, complete the blanks numbered (1) to (12) by inserting the correct terms from the given CSS code. Write each answer in the space provided.



Dropdown Menu inside a **Navigation**

Hover over the "Dropdown" link: **Mission**

Link 3

[1x12=12 Marks]

```
<!DOCTYPE html><html><head><style> /*CSS code*/
ul {list-style-type: none; margin: 0; padding: 0; overflow:
    hidden; background-color: #333;}
li {float: left;}
li a, .dropbtn {display: inline-block; color: white; text-align:
    center; padding: 14px 16px;text-decoration: none; }
li a:hover, .dropdown:hover .dropbtn { background-color: red;
    }
li.dropdown { display: inline-block;}
.dropdown-content {display: none; position: absolute;
    background-color: #f9f9f9; min-width: 160px; box-shadow: 0px
    8px 16px 0px rgba(0,0,0,0.2); z-index: 1; }
.dropdown-content a {color: black; padding: 12px 16px; text-
    decoration: none; display: block; text-align: left; }
.dropdown-content a:hover {background-color: #f1f1f1;}
.dropdown:hover .dropdown-content { display: block;}

</style></head><body> <ul>
<!-- HTML code -->
<li><a href="#"> (1) ----- </a></li>
<li><a href="#"> (2) ----- </a></li>
<li><a href="#"> (3) ----- </a></li>
<li class="(4) ----- ">
<a href="javascript:void(0)" class="(5)----- ">About
    us</a>
<div class="(6) ----- ">
<a href="#">(7)----- </a>
<a href="#">(8)----- </a>
<a href="#">(9)----- </a> </div> </li>
<li><a href="#">Main Products</a></li>
<li class="(10) ----- ">
<a href="javascript:void(0)" class="(11)-----
    ">Main Products</a>
<div class="(12) ----- ">
<a href="#">Link 1</a> </div> </li> </body> </html>
```


II. How does the browser display the outputs of following HTML and CSS code segments?

[a to e:2 x5=10 Marks and f: 3Maks]

- a. `<table border ="1"> <tr>`
`<td colspan=2 rowspan=2>1</td> .....`
`<td colspan=2> 3</td> <td>5</td> .....`
`</tr><tr><td>8</td> <td>9</td> .....`
`<td rowspan=3>10</td> </tr> .....`
`<tr> <td>11</td> .....`
`<td colspan=3 rowspan=3>12</td> .....`
`</tr> tr> <td>16</td></tr> .....`
`<tr> <td rowspan=2>21</td> .....`
`<td>25</td></tr><tr><td>27</td> .....`
`<td colspan=2>28</td> <td>30</td> .....`
`</tr> </table>`
- b. `<html> <head> <style>`
`* { border-style: solid; } .....`
`h1 { border-style: dashed;} .....`
`</style></head><body> .....`
`<h1>Sri Lanka</h1> .....`
`<h2>Sri Lanka</h2> .....`
`</body></html>`
`.....`
- c. `<html><head><style>`
`.Container *{ .....`
`border-style: solid;} .....`
`</style></head><body> .....`
`<div class=Container><h1>UOC</h1> .....`
`<h2>UCSC</h2></div> .....`
`<h4>Sri Lanka</h4></body></html>`
`.....`
`.....`
- d. `<html> <head> <style>`
`div + p {border-style: dashed;} .....`
`</style> </head> .....`
`<body><div> .....`
`<p>Paragraph 1.</p> .....`
`<p>Paragraph 2. </p> </div> .....`
`<p>Paragraph 3.</p> .....`
`<p>Paragraph 4. </p> .....`
`</body> </html>`
`.....`


```
e. <html><head><style>
    [class$="col"]
    {border: 4px solid; color: black;}
</style></head><body>
    <h1>Colombo</h1>
    <div class="firstcol">Colombo.
</div> <br />
    <div class="colmcol">Kandy.</div>
    <div class="start">Galle.</div>
    <p class="mystrcol">Nugegoda.</p>
</body></html>
```

```
f. <html><head><style>
    body {counter-reset: section;}
    h1 {counter-reset: subsection;}
    h1::before { counter-increment: section;
content: "Section " counter(section) ". ";}
    h2::before { counter-increment: subsection;
content: counter(section) "." counter(subsection) " ";}
</style></head><body> <h1>HTML/CSS Tutorials</h1>
    <h2>HTML</h2><h2>CSS</h2>
    <h1>Scripting Tutorials</h1>
    <h2>JavaScript</h2><h2>jQuery</h2></body></html>
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Question 3

I. How does the browser display the outputs of following JavaScript code segments?

[2x2=4 Marks]

a. `let a = 10, b = "10";
document.write("Comparison Operators:
");
document.write("a == b → " + (a == b) + "
");
document.write("a === b → " + (a === b) + "
");
document.write("a > 5 → " + (a > 5) + "
");
document.write("a < 5 → " + (a < 5) + "

");`

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b. `let p = true, q = false;
document.write("Logical Operators:
");
document.write("p && q → " + (p && q) + "
");
document.write("p || q → " + (p || q) + "
");
document.write("!p → " + (!p) + "

");`

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II. How does the browser display the outputs of following PHP code segments?

[2x9=18 Marks]

a. `<?php
$x = "5"; $y = 10; echo $x + $y. " "; echo $x. $y; ?>`

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b. `<?php
$a = [10, "key" => 20, 30]; print_r($a); ?>`

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.....

c. `<?php
$x = 5; echo $x++ . " " . ++$x . " " . $x . "\n"; ?>`

.....
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- d. <?php
 \$nums = [5, 1, 4, 2]; sort(\$nums); print_r(\$nums);
 echo "
"; rsort(\$nums); print_r(\$nums); ?>

- e. <?php
 \$x = 1; while (\$x < 10) {echo \$x; if (\$x == 5) break; \$x++;}
 ?>

- f. <?php
 \$nums = [1, 2, 3]; foreach (\$nums as \$n) {\$n *= 2;}
 foreach (\$nums as &\$n) {\$n *= 2;} print_r(\$nums);echo"
";
 \$n = 100;print_r(\$nums); ?>

- g. <?php
 \$a = [[1,2],[3,4]]; echo \$a[0][1] + \$a[1][0]; ?>

- h. <?php
 \$array = [1, 2, 3, 4, 5];
 \$iterator = new ArrayIterator(\$array);
 \$count = iterator_count(\$iterator);
 echo "Number of elements: \$count"; ?>

- i. <?php
 \$a = ["x" => 1, 2 => 2, 0 => 3, "2" => 4];
 foreach (\$a as \$k => \$v) {echo "(\$k:\$v)";}
 ?>

III. Write down whether the following configurations originate from the Apache or PHP configuration files in the provided spaces to the right.

[1x 3 = 3 Marks]

No	Settings	Apache or PHP
(i)	ServerRoot	
(ii)	DocumentRoot	
(iii)	upload_max_filesize = 3M	

Question 4

Scenario

Assume that you are managing a Student Enrollment System with three tables: `students`, `courses`, and `enrollments`. A student can enroll in multiple courses. The web app runs on XAMPP with the following structure:

```
C:\xampp\htdocs\uoc\index.php
C:\xampp\htdocs\uoc\error\error.php
C:\xampp\htdocs\uoc\con\data\value.php
C:\xampp\htdocs\uoc\con\con.php
C:\xampp\htdocs\uoc\admin\user\logout.php
```

Environment details (subject to change):

- Web server IP: 198.16.10.22
- Client (your PC) IP: 198.168.1.24
- Database server IP: 198.16.10.23
- MySQL root password: ucsc
- Database name: stu
- Tables: `students`, `courses`, `enrollments`

Table specifications:

- `students(student_id, name, email, reg_date)` — primary key auto-increment; use appropriate data types.
- `courses(course_id, course_name, credit)` — primary key auto-increment; use appropriate data types.
- `enrollments(enroll_id, student_id, course_id, enroll_date)` — primary key auto-increment; use appropriate data types; identify relevant foreign keys.

Based on the given details above, answer the following questions.

- I. Write the SQL to create the **`students`** and **`enrollments`** tables with appropriate data types and foreign key constraints. (You may assume `courses` exists.)

[6 Marks]

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- II. The file “con.php” should establish a database connection using the “\$conn” variable and show an error on failure. Write the complete PHP code for “con.php”.

[3 Marks]

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- III. The file “value.php” stores configuration values used by “con.php”. Write the complete PHP code for “value.php”.

[3 Marks]

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- IV. Write PHP code to insert a new student dynamically using a safe (prepared) SQL statement, leveraging “con.php” and “value.php”.

[6 Marks]

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V. Prepare a safe SQL statement to update any student's email dynamically.

[3 marks]

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VI. Write PHP code to delete any course dynamically using a safe (prepared) SQL statement and a MySQLi OOP connection stored in "\$conn".

[4 marks]

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